Norwich Source Protection: Interim Design Review

Presented by UVM Capstone Team:

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Introduction

- UVM Capstone team:
  - Robert Gibson – Environmental Engineering
  - Jessica McGovern – Civil Engineering
  - Michael Perrone – Environmental Engineering
  - Ella Xu – Unspecified Engineering
- Course instructor: John Lens, PhD, PE
- Mentor: John Willis, PE
- Community partner: Mike Wells, Norwich Fire District

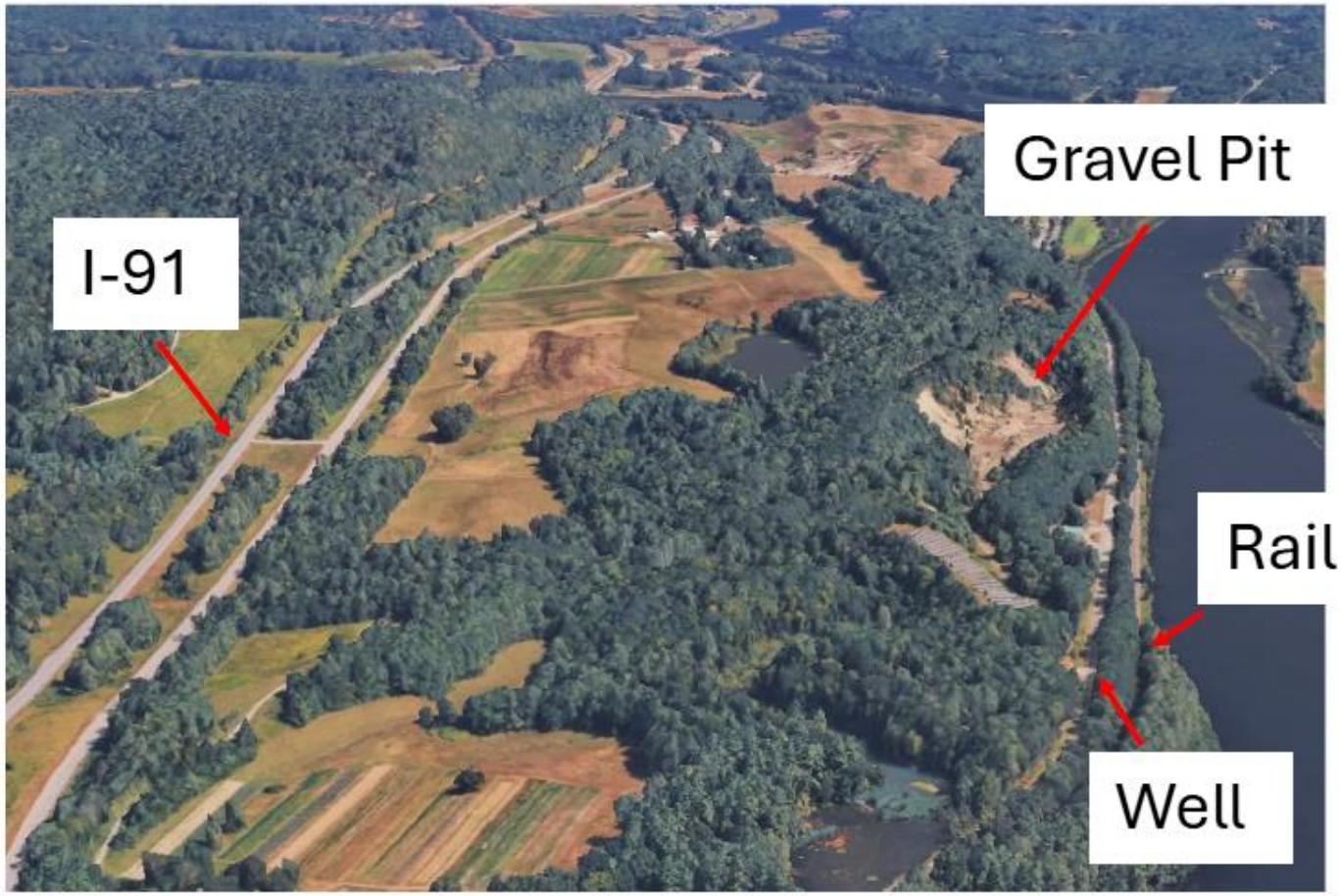
Problem

Problem:

- Gravel pit is changing ownership
- Unaddressed risk from transportation corridors
- Potential contamination from:
  - De-icing materials
  - Vehicle spills

Needs:

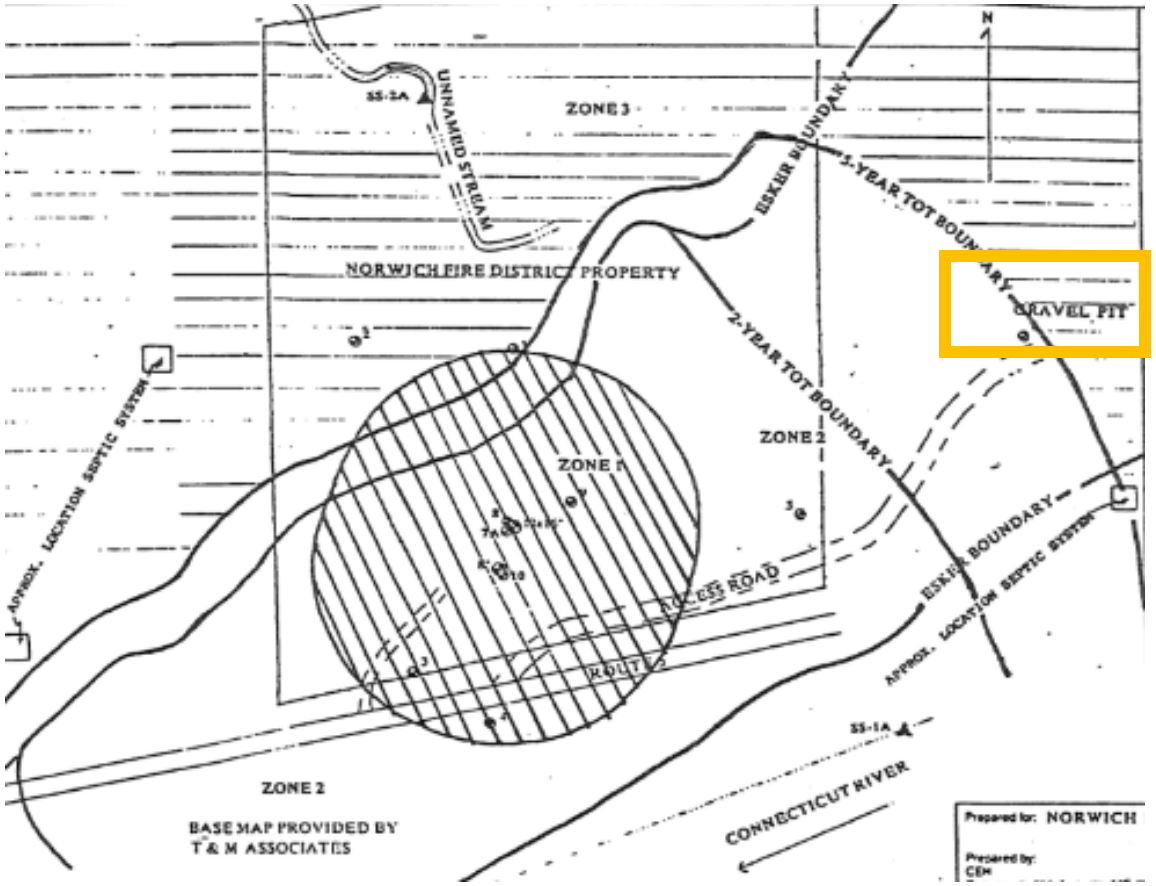
- Risk assessment of land use change
- Updated risk assessment of transportation corridors
- Safe drinking water



Hydrogeologic Setting

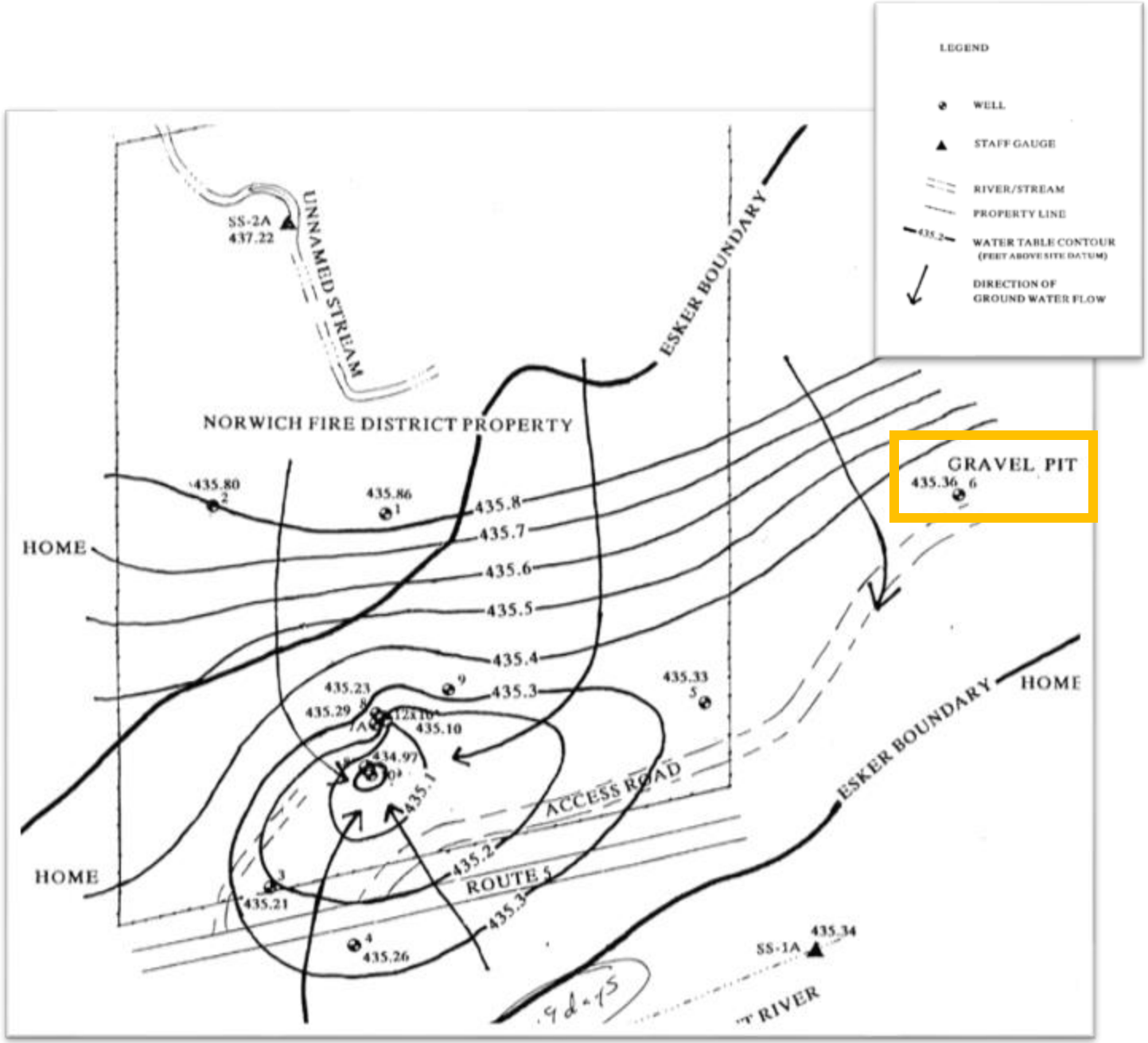
- Area surrounding well has steep embankments
- Surface waters flow toward the unnamed stream, and then towards the Connecticut River
- Well drained soils- sand and gravel
- The well pumps 80% of its water from river groundwater, 20% from the aquifer
- Source protection area divided into three zones





Defining Risk

- Gravel pit sits upstream of the esker between 5 and 6-year travel timeline
- Fuel/oil and deicing chemicals do not break down as they travel downstream and current treatments do not address them



Sources: Caswell, Eichler and Hill hydrogeologic evaluation, 1990. NFD Source Protection Plan, 2025.

Stakeholders & Needs

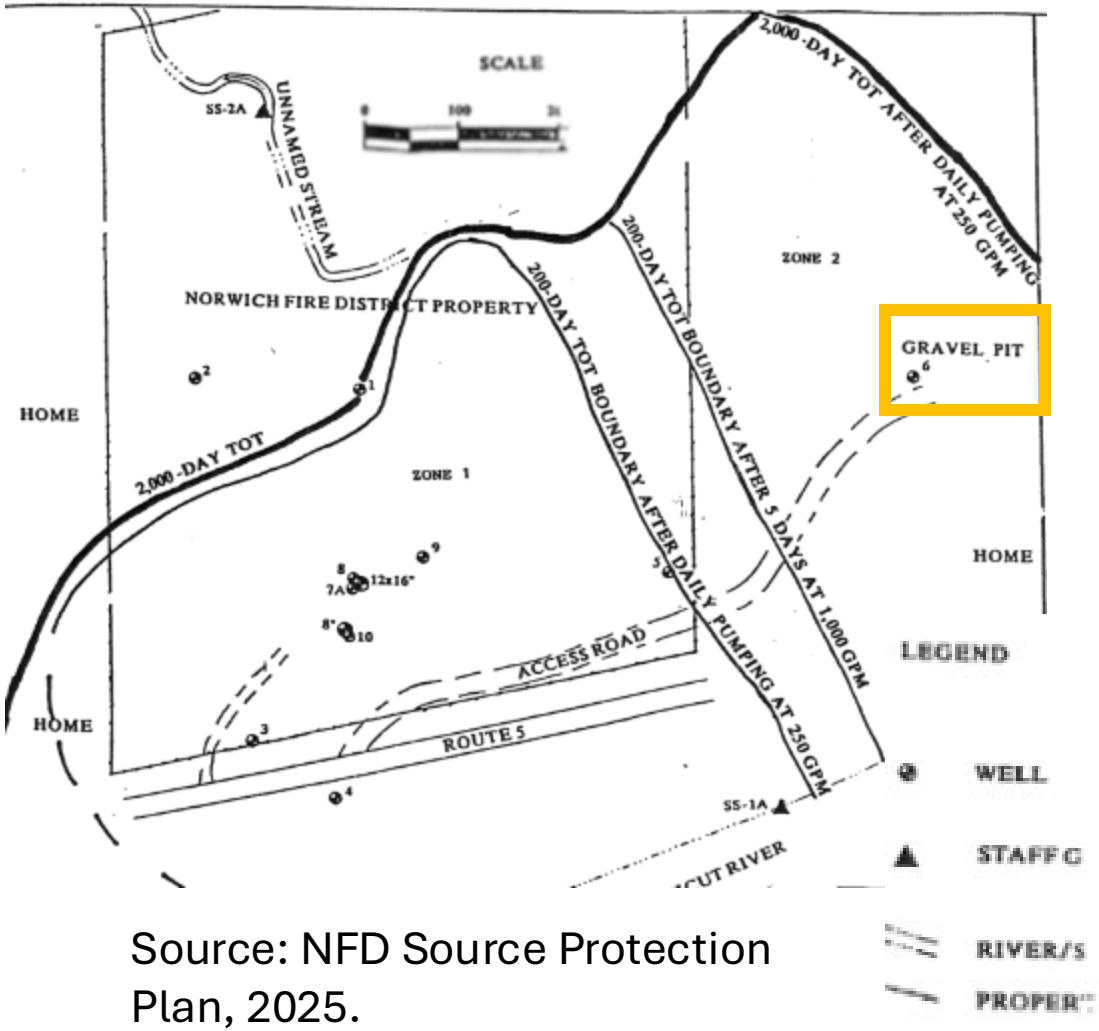
Stakeholder	Relevance
Norwich Fire District	Own and run the water system. Need safe water supply, control of adjacent parcel, affordable solution.
Ratepayers/customers	Need safe, reliable drinking water at affordable rates.
Norwich Town	Enforce zoning rules and project review.
VT DEC/ANR	Have rules for drinking water protection plans.
Norwich Associates	Selling the adjacent parcel containing gravel pit. Interested in fair sale.
Potential buyers of gravel pit	Equipment/material storage site for excavation-related artifact.

Evaluation Criteria

Criterion	Relevance
Risk mitigation	Solution prevents contaminants from reaching drinking water supply
Time scale	Solution continues to work for decades. Doesn't rely on a specific landowner.
Cost	Solution is affordable for NFD and customers.
Feasibility	Solution can be implemented with time, resources, and labor that is available.
Added benefits	Solution provides benefits beyond source protection. (ie. Potential future well site, better data/knowledge, ecological benefits)

Alternative 1: Do Nothing

- Contaminants enter ground in either Zone 2 or 3, undetected
- Mitigation rather than prevention



Source: NFD Source Protection Plan, 2025.

Alternative 1: Do Nothing

- Risks:
  - Does not address US-91 or other transport corridors
  - Relies on reports of contamination- otherwise detected at treatment
  - Environmental harm
  - Increased treatment costs
  - Clean-up costs
    - Including determining source and volume of contamination
  - Customer dissatisfaction
- Benefits:
  - **No immediate costs to NFD**
  - No further human interference to land
  - Contaminants enter in SPA Zones 2 or 3 – time for mitigation (~2000 days)

Alternative 2: Above-Ground Physical Barrier

- Permit excavation company to purchase land
- Company instructed to install an impervious surface for equipment to be stored upon
- Easy cleanup of leaks mitigates contamination risk

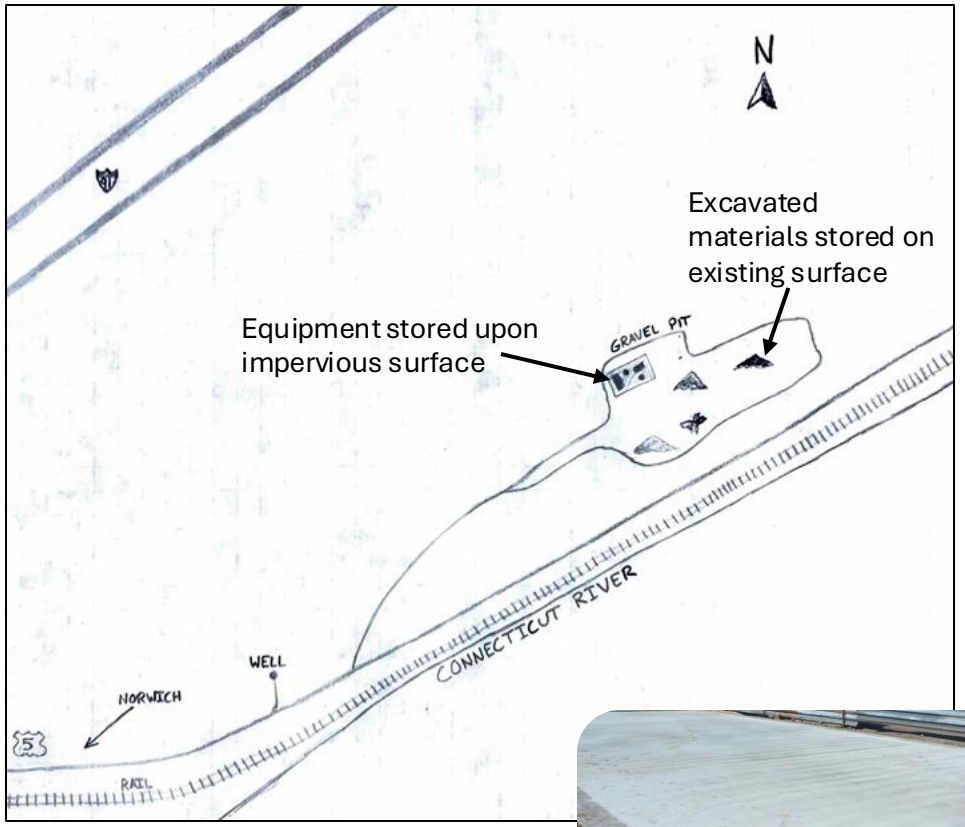




Image Source: Slab Foundation: Types and Relevant Applications - Construction How

Alternative 2: Above-Ground Physical Barrier

- Risks:
  - Does not address US-91 or other transport corridors





  - Relies on reports of contamination or detection at treatment
  - Potential increased treatment costs if contamination occurs
  - Benefits dependent on timely and complete clean up in event of spill

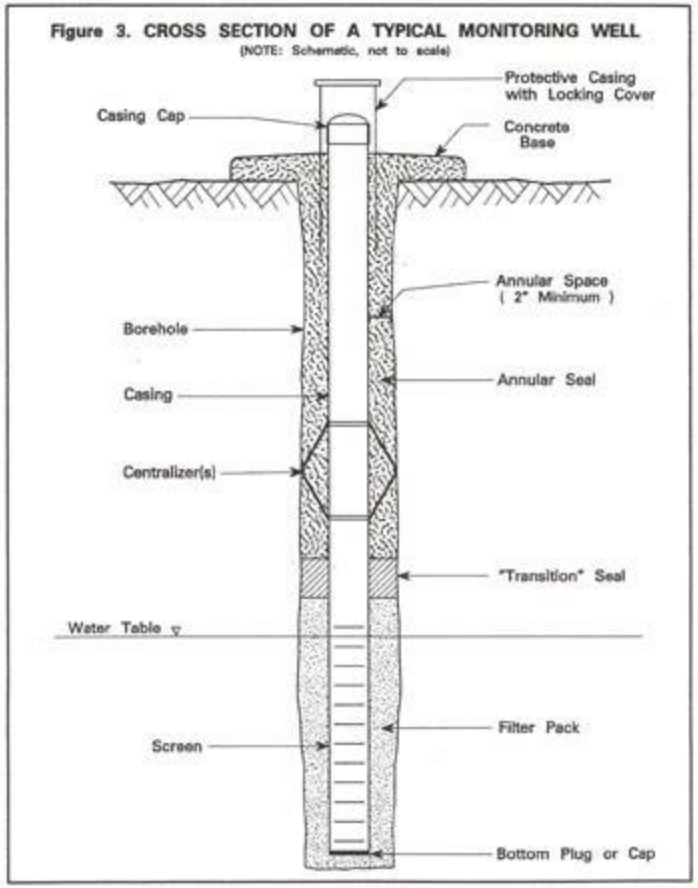
- Benefits:
  - No immediate costs to NFD, unless NFD choses to install the impervious pad themselves
  - Mitigates equipment leakage risks



Image Sources:
Vermont - Interstate 91 Northbound | Cross Country Roads
Vermont Rail System | Flickr
https://storage.googleapis.com/dhgscmvljtzgue/how-to-clean-fuel-oil-off-concrete.html

Alternative 3: New Monitoring Wells

- Permit excavation company to purchase land
- Install 2+ GW monitoring wells; monitor annually (or more freq.)
- Cost: approx. $2k per well (+ annual labor)



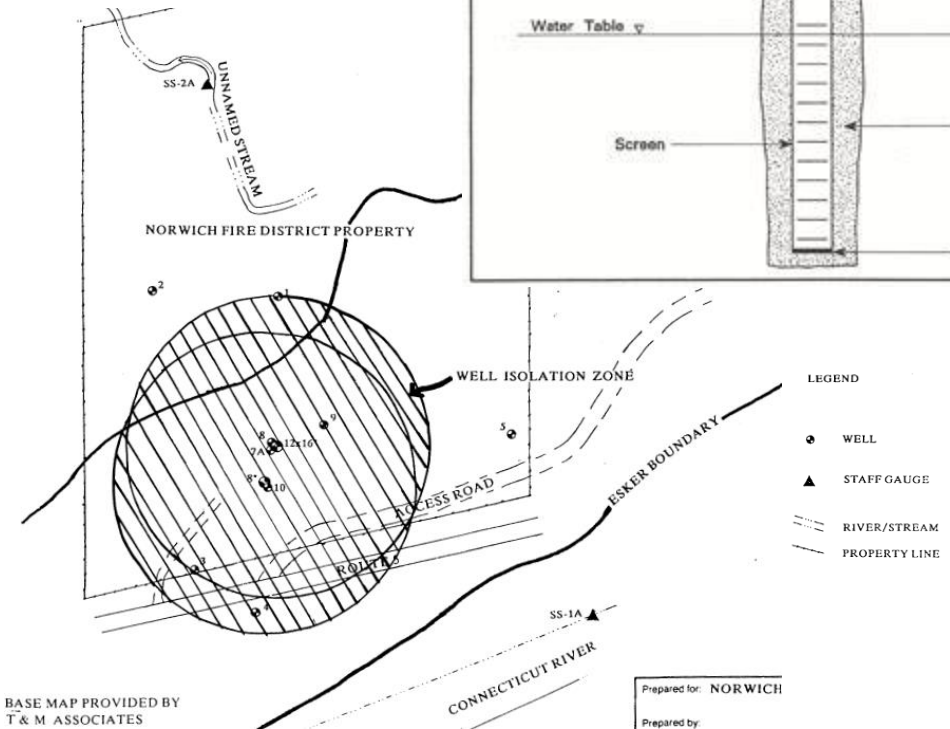


Image Sources:
CA Dept. of Water Resources
Caswell, Eichler and Hill hydrogeologic evaluation, 1990

Alternative 3: Monitoring Wells

- Risks:
  - Does not address contamination if it occurs, only detects if it is being transported towards the wells
  - Requires continued periodic monitoring to be effective over time



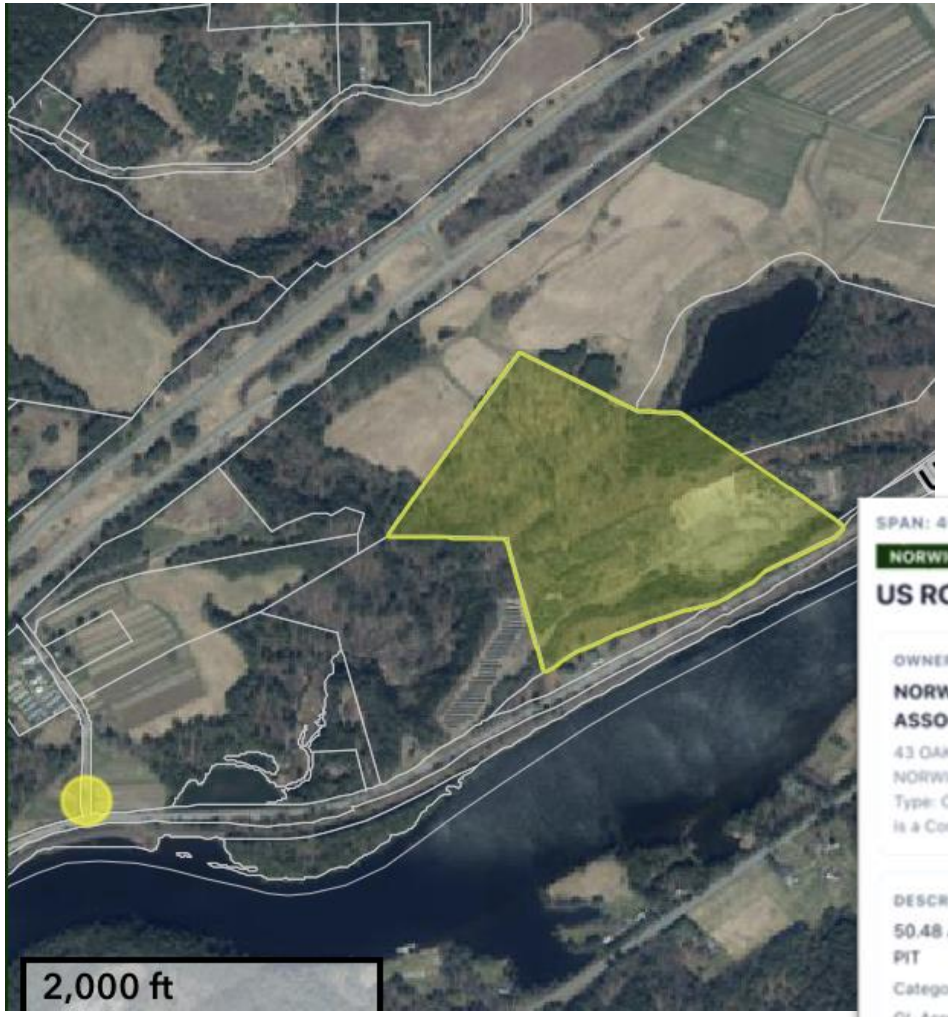
- Benefits:
  - Allows for purchase and use of gravel pit
  - May provide additional info regarding contamination (i.e. from the Connecticut river)
  - Feasible at a low to moderate cost

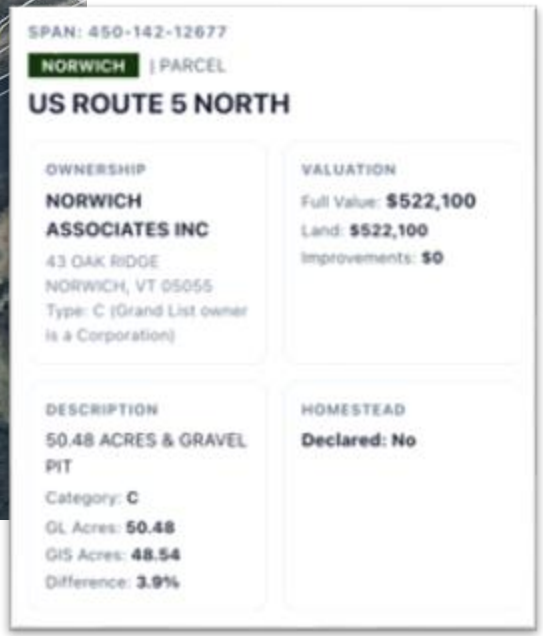


Image Sources:
Walden Environmental Engineering
Timber Homes, LLC

Alternative 4: Buy and Conserve the Land

- NFD purchases gravel pit parcel
- Gravel pit is reclaimed, revegetated, and left undeveloped
- A land trust holds conservation easement; NFD keeps ownership
- monitoring





Source: VT Parcel Viewer App PRD v4.1

Alternative 4: Buy and Conserve the Land

- Pros
  - Removes gravel pit land-use risk instead of managing it
  - Permanent and reliable
  - possible future well site
- Cons
  - High upfront cost
  - Competing with another buyer
  - Does not address possible transportation corridor contamination risk
  - NFD assumes stewardship responsibility

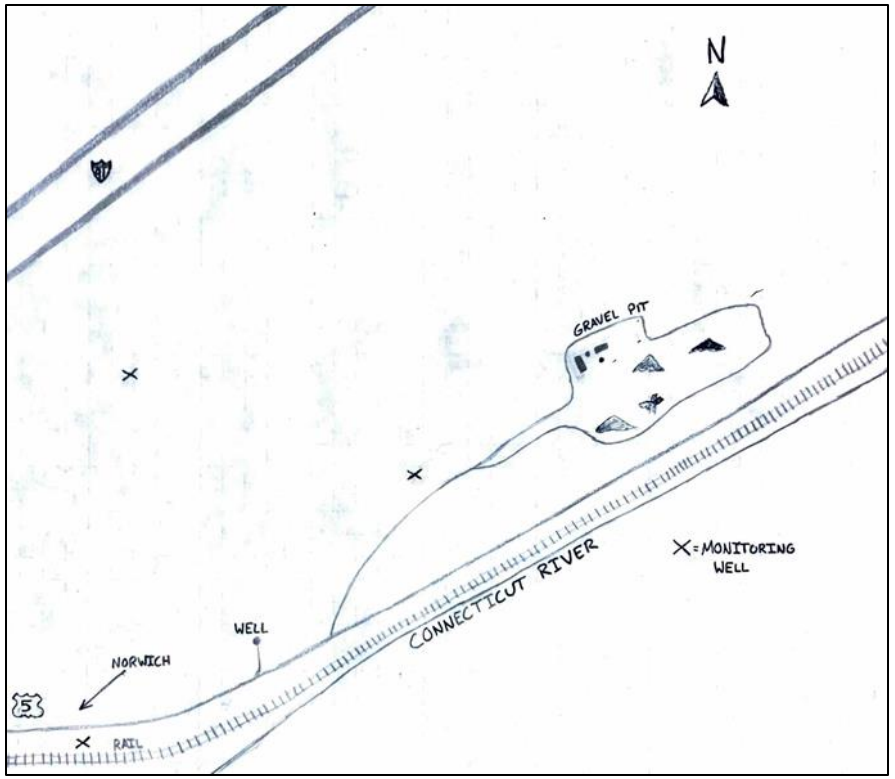
Evaluation Matrix

Criterion	Do Nothing	Impervious surface	Monitoring wells	Buy & Conserve Land
Risk mitigation	0	+	+	+
Time scale	0	+	+	+
Cost (is it affordable?)	0	-	-	-
Feasibility	0	+	+	-
Added benefits	0	0	+	+
Sum	**0**	**2**	**3**	**1**

Preliminary Recommended Solution

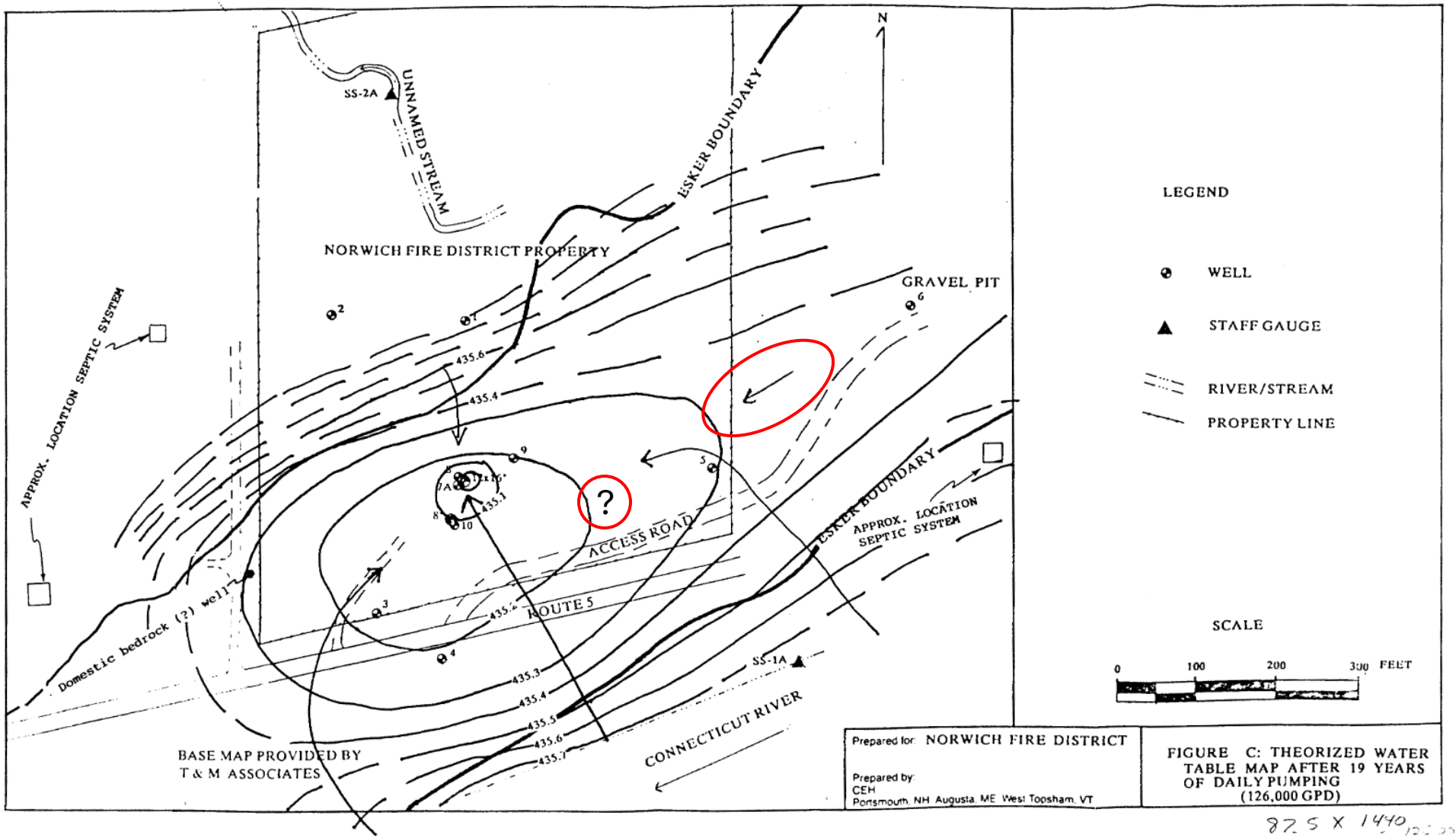
Monitoring wells

- Greatest tool for risk management
- Moderate cost
- Allows for purchase to proceed
- Low maintenance
- Likely best to incorporate into all designs



Example of how monitoring wells may encircle the well

Preliminary Recommended Solution



87.5 X 1440

Questions?